

[Insert Company/Project Logo Here]



Smart Water Meter

Product Manual

Product Version: 1.0
Document Version: 1.0

Prepared by:
Team Excalibur
Department of Electronic and Telecommunication Engineering
University of Moratuwa

Revision History

Version	Date	Description of Change	Author
1.0	July 11, 2026	Initial Release of Product Manual	Development Team

Contents

Revision History	1
1 Introduction	3
2 Product Overview	3
3 Safety Information	4
4 Package Contents	4
5 Product Components	4
6 Internal Hardware Overview	4
7 Technical Specifications	5
8 Installation	6
9 Device Operation	6
10 LED Indicators	6
11 Maintenance	6
12 Troubleshooting	7
13 Frequently Asked Questions	7
14 Warranty	7
15 Regulatory & Safety Information	7
16 Appendix	8

1 Introduction

1.1 Purpose

The Waterboard Smart Water Meter is designed to automatically monitor, record, and securely transmit water consumption data. It eliminates the need for manual meter reading by providing near real-time telemetry to the centralized Waterboard Management System.

1.2 Scope

This manual covers the installation, operation, safety guidelines, and basic maintenance of the Smart Water Meter. It provides an overview of the internal hardware for service technicians but does not replace the comprehensive *Hardware Maintenance Manual* required for board-level repairs.

1.3 Intended Users

- **Customers:** For basic operation and understanding of the local display.
- **Water Board Technicians:** For field installation and troubleshooting.
- **Maintenance Personnel:** For diagnostic reviews and battery replacement.

2 Product Overview

The Smart Water Meter is an ultra-low-power IoT edge device engineered for extreme longevity and precision metrology.

Key Features:

- **Smart water consumption monitoring:** High-precision measurement using internal sensors.
- **Automatic hourly GSM data transmission:** Secure telemetry updates to the central billing server.
- **Long-life battery operation:** Designed for multi-year deployment using industrial-grade lithium cells.
- **Remote billing support:** Seamlessly integrates with the admin portal for automated invoicing.
- **Leak and tamper detection:** Continuously monitors for abnormal continuous flow or physical interference.

[Placeholder: Photograph or Rendered Image of the Product]

3 Safety Information

WARNINGS

- **Do not open the enclosure:** Breaking the factory seal exposes sensitive internal electronics to the environment.
- **Do not immerse the device in water:** The device enclosure is not currently certified for submersion. Keep the electronic housing clear of standing water.
- **Installation must be performed by qualified personnel:** Only authorized Water Board technicians should splice or fit the meter into active plumbing lines.
- **The battery must only be replaced by authorized service personnel:** Improper handling poses a severe fire risk.
- **Do not short-circuit, recharge, or puncture the battery:** The internal cell contains highly reactive materials.

Battery Information

- **Battery Type:** Lithium Thionyl Chloride Battery (Li-SOCl₂), Model ER34615M, 3.6 V.
- **Disposal:** Dispose of depleted batteries strictly according to local environmental and hazardous waste regulations. Do not dispose of in standard municipal waste.

4 Package Contents

- 1x Smart Water Meter Unit
- 2x User Guide (Client and Admin)
- 1x Warranty Card

5 Product Components

[Placeholder: Labelled external view diagram]

- **Water Inlet:** Threaded connection for incoming supply.
- **Water Outlet:** Threaded connection for property supply.
- **LCD Display:** Digital interface for local reading (activates on wake).
- **Wake Button:** Tactile switch to activate the local display.
- **Status LED:** Diagnostic indicator for technicians.
- **Enclosure:** Protective device housing.

6 Internal Hardware Overview

The internal circuitry is optimized for ultra-low quiescent current, heavily relying on sleep states, high-efficiency regulators, and isolated power switching for the cellular radio.

6.1 PCB Photograph & Layout

Refer to Appendix A and B for full-scale schematics and layouts.

6.2 Circuit Breakdown

The main logic board is built using the following datasheet-verified components:

- **Microcontroller (MCU - U30): STM32L073CZT6.** An ultra-low-power ARM Cortex-M0+ microcontroller. It manages sensor polling, LCD driving, and deep-sleep cycles. It features integrated RTC capabilities essential for the timed transmission architecture.
- **High-Current Boost Converter (U21): TPS61023DRLR.** A synchronous boost converter capable of delivering up to 3A switching current. Paired with a 2.2uH inductor (L20) and a bank of four 22uF ceramic capacitors (C20-C23), this circuit steps up the ER34615M battery voltage to provide the massive, high-current transient bursts required by the GSM module during transmission.
- **Low-Dropout Regulator (LDO - U20): XC6206 Series.** A highly precise, extremely low quiescent current LDO regulator (typically 1uA). It continuously steps down and stabilizes the battery voltage to supply clean logic-level power to the MCU and sensors without draining the battery during sleep cycles.
- **GSM Load Switching (Q40 & Q41):** To prevent the cellular radio from causing parasitic battery drain while idle, power is completely isolated using a high-side load switch. This utilizes an **AO3401A** P-Channel MOSFET (Q40)—chosen for its low on-resistance—driven by an **MMBT3904** NPN Transistor (Q41) acting as a logic-level gate driver controlled by the MCU.
- **Timing & RTC (OSC30):** A dedicated Crystal Oscillator, stabilized by 22pF load capacitors (C30, C31), drives the MCU's Low-Speed External (LSE) clock to maintain strict hourly wake-up schedules.
- **User Interface (SW30):** An **MJTP1230A** SPST tactile switch has been used to reset the MCU.

7 Technical Specifications

Parameter	Specification
MCU	STM32L073CZT6 (ARM Cortex-M0+)
LDO Logic Regulator	XC6206 Series
GSM Boost Converter	TPS61023DRLR (3A Switching)
GSM Load Switch	AO3401A (MOSFET) / MMBT3904 (BJT)
Communication	GSM Module
Main Battery	ER34615M Lithium Thionyl Chloride
Battery Voltage	3.6 V
Flow Sensor	Hall Effect
Transmission Interval	Hourly (Default)
Measurement Unit	Liters

8 Installation

1. **Shut off water supply:** Ensure the main line is completely isolated.
2. **Install meter:** Fit the meter into the plumbing line.
3. **Ensure correct flow direction:** Check the directional arrow embossed on the housing.
4. **Tighten fittings:** Use provided accessories; ensure seals prevent leaks.
5. **Verify GSM signal:** Press the Wake button to ensure power, and check the dashboard.
6. **Activate device:** The meter will automatically enter its sleep cycle.

9 Device Operation

- **Measurement:** Internal sensors continuously monitor magnetic rotation, accumulating Liters.
- **Hourly Transmission:** The RTC wakes the MCU hourly. The MCU signals the MMBT3904 to open the AO3401A MOSFET, supplying power from the TPS61023DRLR boost converter to the GSM module. Once data is sent, the switch is closed.
- **Sleep Mode:** Between transmissions, the device enters deep sleep, drawing minimal power via the XC6206 LDO.
- **Automatic Wake-up:** The tactile wake button triggers a hardware interrupt for local reading.

10 LED Indicators

LED Status	Meaning
Green (ON)	Normal Operation / Wake cycle active
Red (ON)	Power on

11 Maintenance

- **Cleaning:** Wipe the enclosure with a damp, non-abrasive cloth.
- **Visual Inspection:** Regularly check housing and fittings for leaks.
- **Battery Replacement:** The ER34615M Lithium Battery is **not user-serviceable**. Replacement must be performed by authorized service personnel to ensure electrical safety and correct system operation.

12 Troubleshooting

Problem	Possible Cause	Solution
Offline (No Data)	No GSM signal / RF Shielding	Move heavy metal objects away. Verify network.
No display on wake	Battery depleted	Contact service for ER34615M replacement.
Incorrect readings	Sensor issue / Blocked impeller	Contact technical support.
Water leak	Installation fault	Shut off mains, inspect fittings.

13 Frequently Asked Questions

Q: How often is data transmitted?

A: The meter connects to the cellular network and updates the dashboard once every hour.

Q: Can I replace the battery myself?

A: No. To maintain the integrity of the device and ensure safety, replacement must be done by authorized technicians.

Q: What happens if the GSM network is unavailable?

A: The meter stores consumption data locally and transmits the backlog once connection is restored.

14 Warranty

Covered by a standard manufacturer's warranty against defects. **Void:** Opening the device enclosure, unauthorized battery replacement, physical damage, freezing, or tampering.

15 Regulatory & Safety Information

Designed for educational and research purposes.

(Pending commercial environmental testing and certifications).

16 Appendix

Appendix A: PCB Schematics

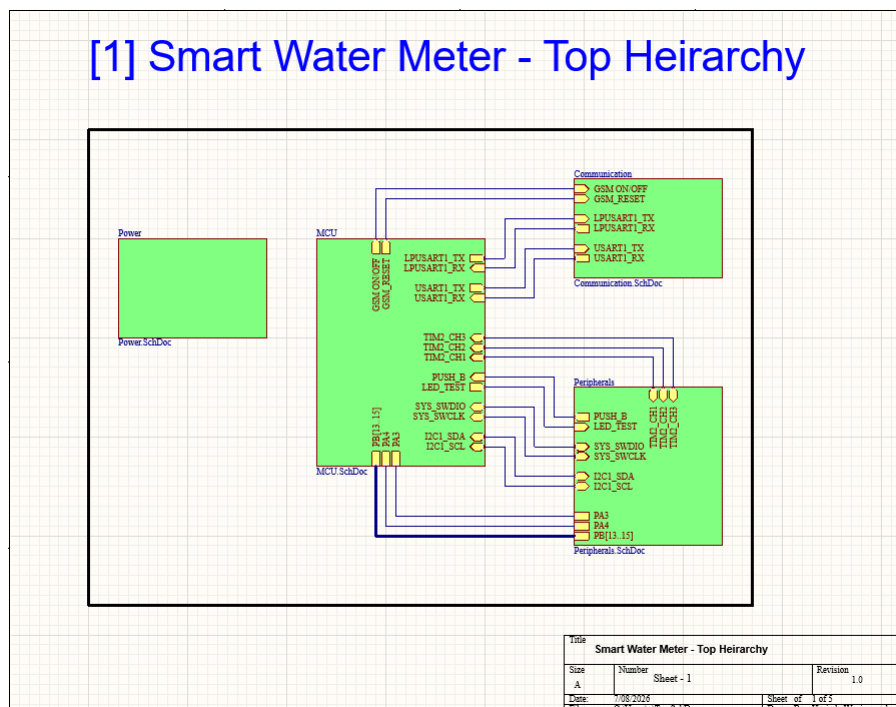


Figure 1: Top connection sheet

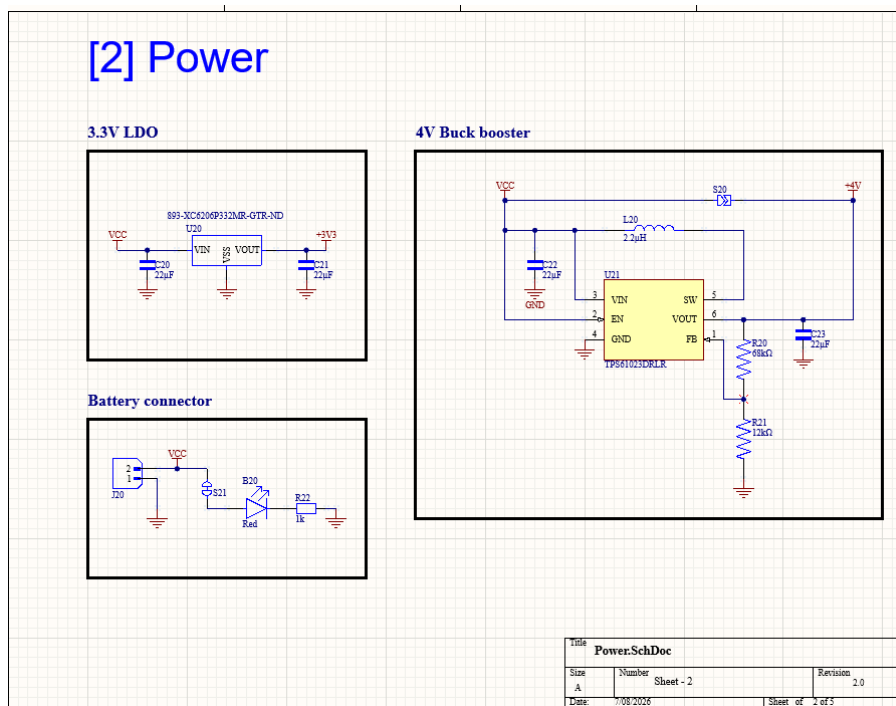


Figure 2: Power sheet

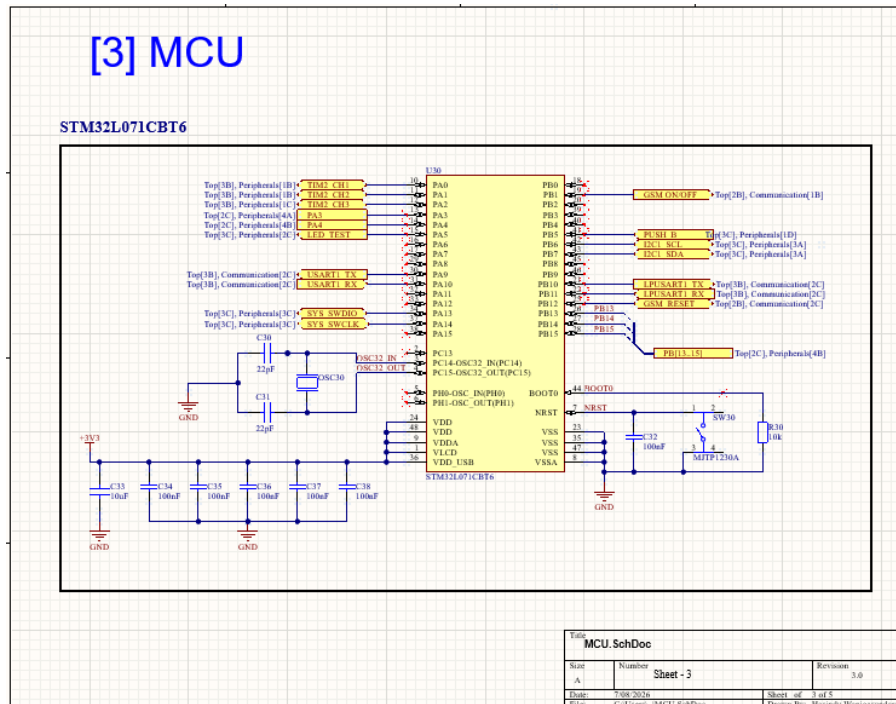


Figure 3: Microcontroller sheet

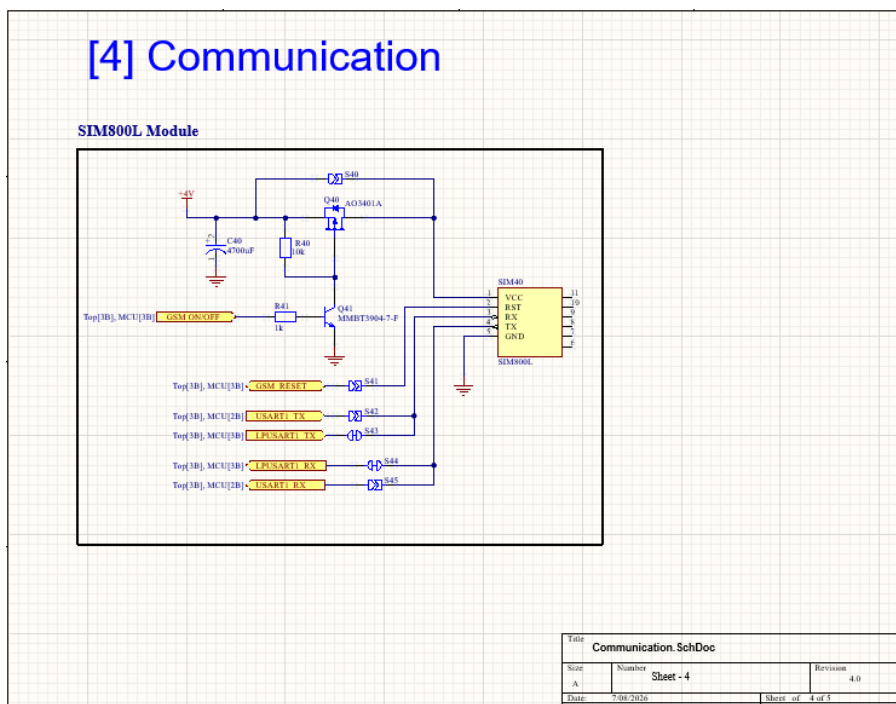


Figure 4: Communication module sheet



Figure 6: PCB-overview (2D-layout)